

Thu, Jun 4 at 10:31 AM

hi as you know i m jobless apst 8 month and looking for data engineering role starting with ebook store project as oltp dataset so we built one prompt so i will give that prompt and you have taught some basic already so we can continue ahead so should i paste the prompt ?

Yes, paste the prompt.

I'll review it and then continue from exactly where your eBook Store Data Engineering project currently stands, without repeating what you've already learned.

Once you paste it, I'll help you with:

- Current project architecture review
- Next learning step in the pipeline
- Python implementation
- Snowflake integration
- ETL/ELT concepts
- Data modeling (OLTP → Analytics)
- Interview-relevant explanations
- Industry best practices

Since your goal is a Data Engineering role after a long gap, we'll treat this project as both:

1. A learning project
2. A portfolio project you can discuss confidently in interviews

Paste the full prompt, and we'll continue from there.



here we go:

the prompt you gave as

<-----start----->

You are my Senior principle Data Engineering Mentor,
System Design and architecture Coach, and Hands-on
Project Guide and tech expert in software engineering.

Show more ▼

Yes, definitely tell me what has already been taught.

Based on the prompt, we should avoid re-teaching things you've already covered.
Otherwise we'll waste time and break the "incremental architecture building" rule.

Give me a short status report like this:

Topic	Status
Business Understanding	Completed / Partial
Data Flow Design	Completed / Partial
OLTP Schema Review	Completed / Partial
PostgreSQL Extraction	Completed / Partial
Python Project Structure	Completed / Partial
JSON Files	Completed / Partial
Parquet Files	Completed / Partial
S3 Simulation Storage	Completed / Partial
Snowflake Setup	Completed / Partial
Raw Layer Design	Completed / Partial
COPY INTO	Completed / Partial
dbt	Completed / Partial
Airflow	Completed / Partial
CDC	Completed / Partial

And if possible, also tell me:

Last thing we built

Example:

- Created extraction pipeline from PostgreSQL
- Exported users/books/orders to JSON
- Added Parquet generation
- Created manifest files
- Created raw folder structure
- Loaded data into Snowflake raw tables
- etc.

Current Folder Structure

Paste only the latest `data_platform/` structure if it has changed since the prompt.

Current Question

Tell me:

"We stopped at _____ and next we planned to learn _____."

Then I'll continue as your Senior Data Engineering Mentor from exactly that point and keep the learning aligned with:

- PostgreSQL → Python → JSON/Parquet → S3 pattern
- Snowflake Raw Layer
- dbt Staging → Intermediate → Marts
- Airflow Orchestration
- Data Quality
- CDC

without resetting the project.



Since you're asking for an updated reusable document, here's the cleaned-up version for your `howtorun.md` .

Edit

Data Platform - End-to-End Execution Guide

1 Insert / Update data in PostgreSQL

Use one of the following:

- Django Admin
- pgAdmin
- SQL Query

Example:

```
UPDATE users_user
SET
    email='new@gmail.com',
    is_active=false
WHERE id=1;
```

2 Extract PostgreSQL → Local JSON

Run the extraction script.

```
python -m scripts.run_extraction
```

Output:

PostgreSQL



Python Objects



ingestion/storage/raw/users_user/YYYY-MM-DD.json

3 Load Local JSON → Snowflake RAW

```
python -m scripts.run_snowflake_loader
```

Output:

Snowflake

BOOKSTORE_DW.RAW.USERS_USER

4 Run dbt Models

```
python transform/run_dbt_run.py
```

Builds:

RAW



STAGING



INTERMEDIATE



MARTS

Creates:

`STAGING.STG_USERS`

`INTERMEDIATE.INT_USERS`

`MARTS.DIM_USERS`

`MARTS.FACT_USERS_INCREMENTAL`

5 Run Snapshot (SCD Type 2)

```
python transform/run_dbt_snapshot.py
```

Builds:

`SNAPSHOTS.USERS_SNAPSHOT`

This preserves historical versions of user records.

6 Run Data Quality Tests

```
python transform/run_dbt_test.py
```

Current tests:

- `not_null`
- `unique`

Future tests:

- relationships
- accepted_values
- freshness

Verify in Snowflake

Check the following objects:

RAW.USERS_USER



STAGING.STG_USERS



INTERMEDIATE.INT_USERS



MARTS.DIM_USERS



MARTS.FACT_USERS_INCREMENTAL



SNAPSHOTS.USERS_SNAPSHOT

8 Update PostgreSQL Again

Example:

```
UPDATE users_user
SET
    email='updated@gmail.com',
    is_active=true
WHERE id=1;
```

or insert a new record.

9 Execute the Pipeline Again

```
python -m scripts.run_extraction
```

```
python -m scripts.run_snowflake_loader
```

```
python transform/run_dbt_run.py
```

```
python transform/run_dbt_snapshot.py
```

```
python transform/run_dbt_test.py
```

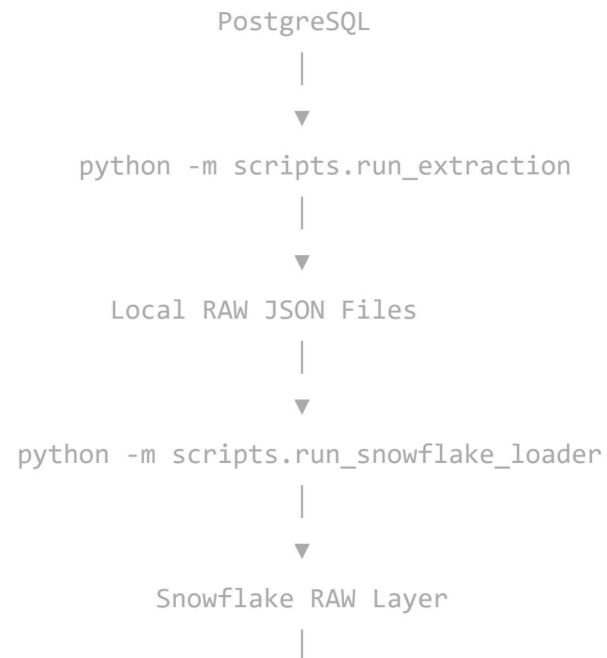
10 Verify Snapshot History

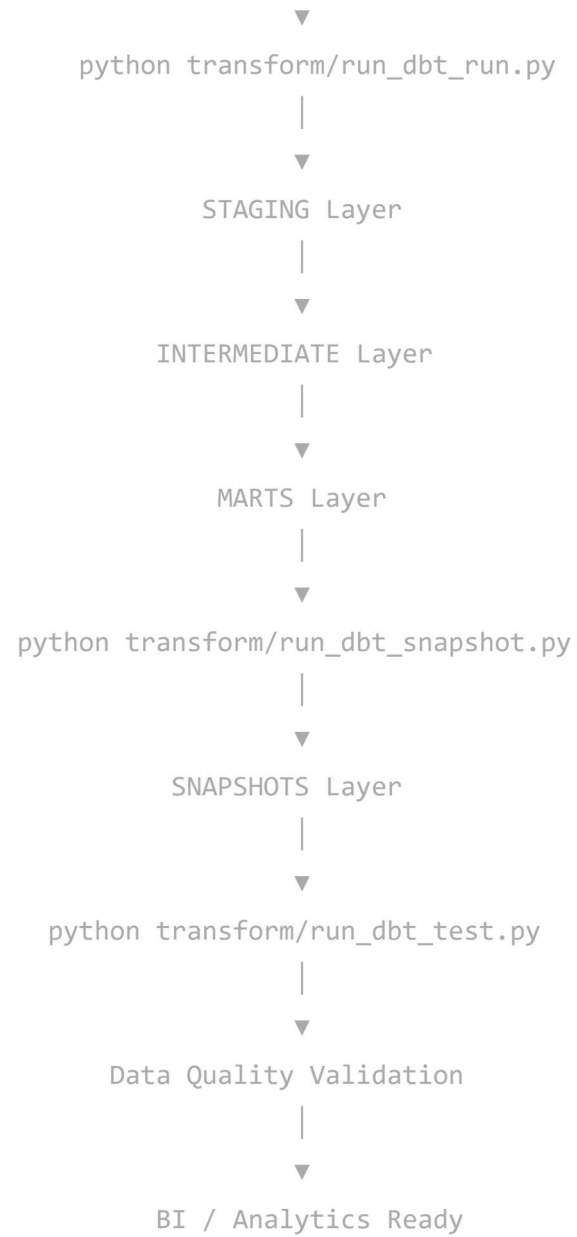
```
SELECT *  
FROM SNAPSHOTS.USERS_SNAPSHOT  
ORDER BY DBT_VALID_FROM DESC;
```

You should see:

- previous version
- latest version
- valid_from
- valid_to

Complete ELT Pipeline Flow





Future Production Flow (Airflow)

In production, these commands are executed automatically by an Airflow DAG:

```
Extract
  ↓
Load RAW
  ↓
dbt Run
  ↓
Snapshot
  ↓
dbt Test
  ↓
Notify / Monitor
```

No manual execution is required once orchestration is configured.

One small correction based on your current setup: if your extraction script **already writes the JSON file** (i.e., it internally calls `LocalLoader`), then **Step 3 becomes the Snowflake loader directly**, and you do **not** need a separate `run_local_loader.py`.

From the code you've shared so far, I suspect that's the case. If you share `run_extraction.py`, I can confirm whether `run_local_loader.py` is actually needed or can be removed entirely.



Fri, Jun 19 at 8:34 AM

so tick those topic whihc wehave covered from this data_platrform book store pipeline project and then we will integrate remaining

topics:

- ❄️ 1.SNOWFLAKE CORE (TOPICS)

❄️ Snowflake

****Easy:**** Architecture, Cloud-native model, Storage vs Compute vs Services, Databases, Schemas, Tables, Warehouses, Roles, Users, Stages, Internal vs External

Show more ▾

For your Bookstore Data Engineering Project, here's an honest production-level assessment.

❄️ 1. SNOWFLAKE CORE

✅ Covered

Easy

- ✅ Snowflake Architecture
- ✅ Database
- ✅ Schemas
- ✅ Tables
- ✅ Warehouses

- ☒ Roles (basic)
- ☒ Internal Stage
- ☒ COPY INTO
- ☒ Auto Suspend/Resume (concept)
- ☒ Credit Basics
- ☒ Query History Basics

Medium

- ☒ Metadata Columns
- ☒ CDC Pipeline (basic metadata-driven)
- ☒ LIST Stage Files
- ☒ REMOVE Stage Files

Hard

- ☐ Streams
- ☐ Tasks
- ☐ Snowpipe
- ☐ Resource Monitor
- ☐ Query Profile
- ☐ Caching
- ☐ Clustering Keys
- ☐ Search Optimization
- ☐ RBAC Design
- ☐ Masking Policies
- ☐ Row Access Policies
- ☐ Snowpark
- ☐ Stored Procedures
- ☐ UDF

- ❌ Multi Cluster Warehouse

SQL

✓ Covered

Easy

- ✓ SELECT
- ✓ WHERE
- ✓ GROUP BY
- ✓ CASE
- ✓ NULL handling
- ✓ JOIN
- ✓ Aggregations

Medium

- ✓ CTE
- ✓ Views
- ✓ Window Functions (basic)

Hard

- ✓ Incremental MERGE logic
- ✓ SCD2 concepts

Still Remaining

- ❌ Recursive CTE
- ❌ Explain Plan
- ❌ Optimizer

-  Materialized Views



Covered

Easy

-  OOP
-  JSON
-  YAML
-  File handling
-  dotenv
-  Modules

Medium

-  Logging
-  Exception handling
-  Config Driven Design
-  CLI scripts

Remaining

-  Async
-  Multiprocessing
-  Memory optimization



Nothing yet.



Covered

Easy

- ☒ Models
- ☒ refs
- ☒ sources
- ☒ dbt run
- ☒ dbt test

Medium

- ☒ Incremental Model
- ☒ Snapshots
- ☒ Bronze/Silver/Gold Layering

Remaining

- ☐ Freshness
- ☐ Variables
- ☐ Hooks
- ☐ Exposures

Hard

- ☒ Macros (generate_schema_name)

Remaining

-  Packages
-  Slim CI
-  Enterprise packages

Airflow

Covered

Easy

-  DAG
-  BashOperator
-  Dependencies
-  Retries
-  Scheduling

Remaining

-  XCom
-  Sensors
-  Variables
-  Connections
-  Branching
-  Pools
-  Dynamic DAG

AWS

Covered

- ❌ None yet

PostgreSQL

Covered

Easy

- ✅ CRUD
- ✅ Tables
- ✅ Schemas

Medium

- ✅ Incremental extraction
- ✅ Joins

Remaining

- ❌ Triggers
- ❌ Stored Procedures
- ❌ JSONB

Streamlit

Covered

- ❌ None

Data Engineering Core

Covered

Easy

- ☒ ETL
- ☒ ELT
- ☒ OLTP
- ☒ OLAP
- ☒ Pipeline
- ☒ Batch

Medium

- ☒ CDC
- ☒ Incremental Load
- ☒ Data Reconciliation
- ☒ Metadata Driven Pipeline
- ☒ Backfill concept

Remaining

- ☐ Schema Evolution
- ☐ Data Contracts

Hard

Covered

- ☒ Metadata Driven Pipeline

Remaining

- ☐ Observability
- ☐ Late arriving data
- ☐ Event Driven



Data Modelling

Covered

Easy

- ☒ Dimension Table
- ☒ PK
- ☒ FK

Medium

- ☒ Star Schema
- ☒ Surrogate Key
- ☒ SCD Basics

Remaining

- ☒ Snowflake Schema

Hard

- ☒ SCD Type 2

Remaining

- ☒ Data Vault
- ☒ Bridge Tables



Kafka

Covered

- ☒ None



DevOps

Covered

-  Git
-  Logging

Remaining

-  CI/CD
-  Monitoring



Testing

Covered

Easy

-  not_null
-  unique

Remaining

-  Freshness
-  Row Count

Medium

-  Great Expectations



Production Operations

Covered

-  Pipeline Debugging

Remaining

- Everything else

 AI for DE

Covered

-  None

Overall Progress

Domain	Progress
Snowflake	50%
SQL	65%
Python	70%
dbt	80%
Airflow	40%
PostgreSQL	55%
Data Engineering	75%
Data Modelling	70%

Domain	Progress
Testing	45%
AWS	0%
Streamlit	0%
PySpark	0%
Kafka	0%
DevOps	25%
AI for DE	0%



Recommended roadmap from here

This is the sequence that will give you the strongest portfolio and interview readiness:

1. ☒ Finish end-to-end pipeline testing (update PostgreSQL → extract → load → dbt → snapshot → tests)
2. ☒ Build a **Customer 360 Streamlit dashboard** on top of your Snowflake marts.
3. ☒ Learn **Snowpark** by querying Snowflake directly from Streamlit.
4. ☒ Add **Airflow using Docker** and orchestrate the full pipeline.
5. ☒ Add **CI/CD** (GitHub Actions for dbt tests and deployment).
6. ☒ Start **PySpark + Databricks** for big data processing.
7. ☒ Learn **AWS** (S3, Glue, IAM, Lambda).
8. ☒ Move to **Kafka + Streaming** for real-time pipelines.

This order builds naturally from your existing project and mirrors how many production data platforms evolve.



Fri, Jun 19 at 5:05 PM

so based on this data book store learning do you want anything to update in resume?

to get shortlisted fast or not need

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Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with hands-on production experience

Show more ▼

Yes—but **not yet**.

Your current resume is actually **ahead of your current project** in several places. If an interviewer asks you to show those implementations, there will be gaps.

Here's a production-level audit.



Can be added NOW (100% truthful)

Project Title

Instead of

ETL Data Warehouse Pipeline

Use

Metadata-Driven ELT Data Platform (PostgreSQL → Snowflake → dbt → Airflow)

This is much stronger.

Update Tech Stack

Current

Python, Airflow, PostgreSQL, SQL, YAML, Streamlit

Change to

Python

PostgreSQL

Snowflake

dbt Core

SQL

YAML

Docker

Airflow (DAG implementation)

Git

Remove Streamlit until you actually build it.

Better bullets

Current bullets are generic.

Replace with something like

- Designed a metadata-driven ELT pipeline to ingest PostgreSQL data into Snowflake using YAML-based configuration.
- Built a Bronze–Silver–Gold architecture using dbt with staging, intermediate, marts, and snapshot layers.
- Implemented incremental loading and Slowly Changing Dimension (SCD Type 2) snapshots for historical tracking.
- Automated data quality validation using dbt tests (not_null, unique) across transformation layers.
- Created reusable Python ingestion modules for extraction, local raw storage, and Snowflake loading.
- Developed Airflow DAGs to orchestrate dbt run, snapshot, and test workflows.
- Containerized the platform using Docker for reproducible local development.

Those bullets match exactly what you've built.

Skills Section

Move these to the top because they're now proven.

Snowflake
dbt Core
Python
SQL
PostgreSQL
Airflow

Docker
Git
YAML

Things to REMOVE for now

These aren't backed by your current implementation.

✗ Streamlit

(until Customer 360 dashboard exists)

✗ Snowpark

(not learned yet)

✗ Great Expectations

You used dbt tests, not Great Expectations.

✗ PySpark

You haven't started it.

✗ GitHub Actions

Not implemented.

Experience

For your real experience at Modulr, keep only things you genuinely did.

For example:

Current

Applied Great Expectations

If you didn't actually do this,

remove it.

Current

Streamlit dashboards

remove unless true.

Current

Snowpark

remove unless true.

Biggest Improvement

Your project deserves a dedicated section.

Instead of

DATA ENGINEERING PROJECTS

make it

FEATURED PROJECT

Then dedicate 6–8 bullets.

Recruiters notice this much more.

What to add after the next milestones

After Customer 360

Add

✓ Streamlit

After Snowpark

Add

✓ Snowpark for Python

After Airflow Docker

Add

☒ Apache Airflow (Docker)

After CI/CD

Add

☒ GitHub Actions

After PySpark

Add

☒ PySpark

After AWS

Add

☒ S3

Glue

IAM

Lambda

Resume Readiness

Current Resume

Rating: ★★ ★★ ★★ ★★ ★★ ★★ ★★ ☆☆ (8/10)

Very strong, but a little optimistic in places.

After aligning it with what you've actually built

Rating: ★★ ★★ ★★ ★★ ★★ ★★ ★★ ★★ ☆ (9.5/10)

Recruiters appreciate resumes that are technically consistent.

After completing:

- ☒ Customer 360 Streamlit
- ☒ Snowpark
- ☒ Airflow Docker
- ☒ CI/CD
- ☒ PySpark + Databricks

Rating: ★★ ★★ ★★ ★★ ★★ ★★ ★★ ★★ ★★ (10/10)

At that point, you'll have a cohesive, end-to-end portfolio that mirrors a real production data platform—from OLTP ingestion to analytics, orchestration, historical tracking, and visualization. That combination is highly compelling for Data Engineer roles.



Today 9:24 AM

just wanted to update how many step we have done and what remaining on data platform project



Book Store Data Platform Progress

You have already completed the majority of what most junior-mid Data Engineers build in portfolio projects.



Phase 1 — Source System (OLTP)

Status: 100% Complete

Completed

-  Django Bookstore application
-  PostgreSQL OLTP database
-  Users table
-  Created_at / Updated_at columns
-  Data generation through application



Phase 2 — Metadata Driven Ingestion

Status: 100% Complete

Completed

-  YAML configuration driven ingestion
-  Metadata-based column selection
-  No `SELECT *`
-  Watermark column configuration
-  Primary key definition
-  Config driven architecture

Files:

```
ingestion/config/users_user.yml  
utils/yaml_reader.py
```

Phase 3 — Extraction Layer

Status: 100% Complete

Completed

-  PostgreSQL connector
-  Metadata driven extractor
-  Dynamic query generation
-  Dictionary output

Files:

```
utils/db_connection.py  
ingestion/extractors/postgres_extractor.py
```

✓ Phase 4 — Raw Landing Storage

Status: 100% Complete

Completed

- ✓ Local raw storage
- ✓ Date partitioned files
- ✓ S3 simulation

Files:

```
ingestion/loaders/local_loader.py  
ingestion/storage/raw/
```

✓ Phase 5 — Snowflake RAW Layer (Bronze)

Status: 100% Complete

Completed:

- ✓ Database creation
- ✓ Schema creation
- ✓ Internal stage
- ✓ RAW table
- ✓ JSON ingestion
- ✓ COPY INTO

Files:


```
warehouse/client.py  
warehouse/ddl/  
warehouse/stages/  
warehouse/loaders/snowflake_loader.py
```

✓ Phase 6 — dbt Setup

Status: 100% Complete

Completed:

- ✓ dbt project
- ✓ profiles.yml
- ✓ schema generation macro
- ✓ Python wrapper scripts

Files:

```
transform/dbt_project.yml  
transform/profiles.yml  
transform/macros/generate_schema_name.sql
```

✓ Phase 7 — STAGING Layer

Status: 100% Complete

Completed:

- ✓ Source declaration
- ✓ Data type standardization

-  Flattening
-  Cleaning

Result:

```
RAW.USERS_USER  
↓  
STAGING.STG_USERS
```

Phase 8 — INTERMEDIATE Layer

Status: 100% Complete

Completed:

-  Business logic
-  Derived columns
-  Email status
-  Active flags

Result:

```
STAGING.STG_USERS  
↓  
INTERMEDIATE.INT_USERS
```

Phase 9 — MART Layer

Status: 100% Complete

Completed:

-  User segments
-  Analytics-ready model
-  Gold layer

Result:

```
INTERMEDIATE.INT_USERS  
↓  
MARTS.DIM_USERS
```

Phase 10 — Data Quality

Status: 100% Complete

Completed:

-  not_null tests
-  unique tests
-  dbt test execution

Remaining:

-  relationships tests
-  accepted_values tests

Phase 11 — SCD Type 2 Snapshot

Status: 100% Complete

Completed:

- ☒ Snapshot configuration
- ☒ Historical tracking
- ☒ dbt snapshot execution

Result:

SNAPSHOTS.USERS_SNAPSHOT

☒ Phase 12 — Incremental Models

Status: 100% Complete

Completed:

- ☒ Incremental materialization
- ☒ Unique key
- ☒ is_incremental()

☒ Phase 13 — Airflow DAG Design

Status: 70% Complete

Completed:

- ☒ DAG file
- ☒ dbt run task
- ☒ snapshot task
- ☒ dbt test task

-  task dependencies

Remaining:

-  Install Airflow
-  Docker Compose
-  Airflow UI
-  Trigger DAG execution

Phase 14 — Streamlit Customer 360

Status: 0%

Planned:



Features:

- Customer profile
- Order history
- Active customers
- Revenue metrics
- Segmentation

Phase 15 — Snowpark

Status: 0%

Planned:

- Snowpark session
- DataFrame API
- Snowflake → Streamlit queries

Phase 16 — CI/CD

Status: 0%

Planned:

- GitHub Actions
- dbt test automation
- Docker build
- Deployment

Phase 17 — PySpark Migration

Status: 0%

Planned:

Move from:

PostgreSQL
→ Snowflake
→ dbt

towards:

- PostgreSQL
- S3
- PySpark
- Snowflake
- dbt

Overall Project Completion

Area	Progress
OLTP Source	100%
Ingestion	100%
Local Landing	100%
Snowflake RAW	100%
dbt Setup	100%
Staging	100%
Intermediate	100%
Marts	100%
Data Tests	100%

Area	Progress
Snapshots	100%
Incremental Models	100%
Airflow Design	70%
Streamlit	0%
Snowpark	0%
CI/CD	0%
PySpark	0%

Overall Completion



≈ 75-80% complete

You are essentially finished with the **core Snowflake + dbt data warehouse track**.

The remaining work is mostly around:

1. Visualization (Streamlit)
2. Orchestration execution (Airflow Docker)
3. CI/CD
4. Big Data processing (PySpark)

Those are more "platform maturity" features rather than core warehouse implementation work.

